



A robot you can 'teach' to pack bread

Rodney Brooks has designed a unique robot that is easy to use, comparatively cheap and suitable for even small manufacturing outfits. Baxter can perform repetitive tasks like bread packing, working safely next to people. It exhibits 'common sense' and is capable of sensing and adapting to its task and environment. For example, let us say Baxter is assembling a small machine and it loses a part. It will visually scan the desk, pick up the piece and start working again—just like we do. The best part is that it involves no programming or costly integration. It is possible to train the robot by physically moving its limbs and demonstrating how it should work. You can easily update its capabilities through software updates, or create your own capabilities with an easy software development kit.

Inside: Seven degrees of freedom, 8-12 pick-and-place operations per minute, 1m/s arm speed, five built-in cameras, interchangeable end-effectors, force sensing, springs and force control at each joint for free movement and bump detection, behaviour-based intelligence, vision-guided movement, human detection using front camera and 360-degree sonar, navigator and face-like display for human-machine interface (HMI), casters for movement, locking feet for stability. Full specifications and parts list available at www.rethinkrobotics.com

A bouncy tool for police encounters

When scouting out suspects, police and military personnel always fear the unknown—how many criminals are inside a room, whether they have any dangerous weapons with them and so on. Similarly, firefighters would like to know how intense a fire is, how many people are stuck in the accident, etc. Bounce Imaging, a throwable electronic device, aims to help in such situations by capturing images and other information using cameras and sensors, and relaying it to a mobile device.

Inside: Still in the design stage, Bounce Imaging is likely to include six wide-angle cameras (that can together give a 360-degree view), each capable of taking two snaps per second and surrounded by infrared LED flash, a protective casing that allows the device to bounce without any breakage of components, a battery, a variety of sensors such as oxygen, smoke, temperature and radiation sensors, and an accelerometer and gyroscope to orient images. The captured data is sent wirelessly to an Android-based mobile device, where an application helps analyse the information. Developers are still working on some solution (perhaps a tether or motion capabilities) to retrieve the device after it has done its job.

